Week 03 teacher guide

Rates and percents

Keep this guide beside PowerPoint slides 73-109. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 015. Saturday | pages 63-67 | slides 73-90
Rates and unit rates

Day 016. Sunday | pages 68-78 | slides 91-109
Percent as a rate per hundred

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Weekly check

This week’s idea

A unit rate describes an amount per one unit. Percent means per hundred. Label the whole before deciding whether to find a part, a whole, or a percent.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Rates and unit rates

Day 015, pages 63-67 | PowerPoint slides 73-90

Goal: I can find an amount per one unit and use it to compare rates.

Set out: ratio table template, double number line template, pencil.

Check first: Equivalent ratios and decimal division by whole numbers.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Labels dollars per apple. Why it matters: The units identify division order. Tutor move: Ask what one in the rate refers to.

Uses scaling or divides to one. Why it matters: Both methods can find an equivalent rate. Tutor move: Accept either, then connect them in a table.

Converts time to 60 minutes. Why it matters: Hourly rates require an hour as the time unit. Tutor move: Mark 0, 15, 30, 45, 60 on the time line.

Likely misconceptions

Divides in the reverse direction. Why: Does not track the per unit. Tutor move: Read the desired rate aloud before dividing.

Treats 20 minutes as 0.20 hours. Why: Uses base ten instead of 60 minutes per hour. Tutor move: Partition an hour into three 20-minute sections.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 73. Week 3, Saturday

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: I can find an amount per one unit and use it to compare rates.

Ask: What do you already know about this idea?

Expect: Equivalent ratios and decimal division by whole numbers.

If stuck: Ask what one in the rate refers to.

Slide 74. Today

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.

If stuck: Read the learning goal together.

Slide 75. Recall | 4 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{11}{12}$; 2: 8 yellow; 20 purple; 3: $\frac{1}{6}$ liter; 4: 3 cups

If stuck: Give time to draw before giving a hint.

Slide 76. Recall: answers | 3 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{11}{12}$; 8 yellow; 20 purple; $\frac{1}{6}$ liter; 3 cups

If stuck: Rebuild one missed item with its model.

Slide 77. Launch | 8 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide.

Expect: P1: 5 kilometers per hour

If stuck: Ask what one in the rate refers to.

Slide 78. Explore | 4 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide.

Expect: P2: \$0.75 per apple; P3: $\frac{2}{5}$ cup per serving

If stuck: Ask what one in the rate refers to.

Slide 79. The model | 4 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Four 15-minute intervals make one hour.

If stuck: Ask what one in the rate refers to.

Slide 80. Required practice | 3 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide.

Expect: P4: 5 cards per minute; $\frac{1}{5}$ minute per card; P5: Pack B: $0.35 per pencil

If stuck: Ask what one in the rate refers to.

Slide 81. Required practice | 3 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide.

Expect: P6: 9 miles per hour; P7: 0, 8, and 20 meters

If stuck: Ask what one in the rate refers to.

Slide 82. Watch the thinking | 3 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 4 miles per hour

If stuck: Return to the matching unit in the model.

Slide 83. Your turn | 3 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 2; 10 miles per hour

If stuck: Compare with the previous example without copying its numbers.

Slide 84. Compare methods | 8 min

Topic: Rates and unit rates

Student: Day 015, pages 63-67

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide. Partner compare: What does $\frac{1}{5}$ minute per card mean compared with 5 cards per minute?

Expect: Accept drawings and equations that preserve the same quantities and units. Uses scaling or divides to one.

If stuck: Accept either, then connect them in a table.

Slide 85. A claim to check | 7 min

Topic: Rates and unit rates

Student: Day 015, pages 63-67

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: No. \$6 ÷ 8 apples = \$0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

If stuck: Write dollars above the division and apples below it.

Slide 86. Words and meaning | 4 min

Topic: Rates and unit rates

Student: Day 015, pages 63-67

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: 15 kilometers in 3 hours. 5 kilometers per hour. \$0.60 per apple means \$0.60 for one apple.

If stuck: Read one definition and draw its example.

Slide 87. In your own words | 3 min

Topic: Rates and unit rates

Student: Day 015, pages 63-67

Say: Use your own words and one example.

Ask: Agree or disagree: A unit rate must be a whole number. Use the apples to decide.

Expect: A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.

If stuck: Point to the model and name what each part represents.

Slide 88. Exit ticket | 8 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 12 miles per hour; 5 minutes per mile

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 89. The week ahead | 5 min

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 90. Tutor reference

Topic: Rates and unit rates
Student: Day 015, pages 63-67

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 12 miles per hour; 5 minutes per mile

Got it: Gives 12 miles per hour and 5 minutes per mile, with correct units and scaling.

Almost there: Gets one rate correct or both values correct with one unclear unit.

Needs re-teaching: Reverses division, uses 45 as 0.45 hours, or assigns the rates to the wrong units.

Sunday. Percent as a rate per hundred

Day 016, pages 68-78 | PowerPoint slides 91-109

Goal: I can find a percent, a part, or a whole using matching amounts.

Set out: percent strips, double number line template, pencil.

Check first: Equivalent ratios and unit rates.

Pacing: Retrieval 4; launch 4; explore 12; discuss 10; synthesize 3; Mini Check A 15; exit 5; preview 5; slack 7. Total 65 min.

Explore: what to watch for

Labels the given amount as part or whole. Why it matters: The role controls the calculation. Tutor move: Ask which amount belongs at 100%.

Combines benchmark percentages. Why it matters: Builds a non-benchmark percent from known pieces. Tutor move: Accept a unit-percent method or a benchmark method.

Allows a percent above 100. Why it matters: A comparison amount can exceed the starting whole. Tutor move: Extend the percent line beyond 100%.

Likely misconceptions

Multiplies the known part by the percent. Why: Treats every percent problem as finding a part. Tutor move: Place the known part and its percent on aligned lines.

Rejects 125%. Why: Thinks a percent is always a piece below one whole. Tutor move: Show one whole plus one fourth of the same whole.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 91. Week 3, Sunday

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: I can find a percent, a part, or a whole using matching amounts.

Ask: What do you already know about this idea?

Expect: Equivalent ratios and unit rates.

If stuck: Ask which amount belongs at 100%.

Slide 92. Today

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Percent compares an amount with a whole split into 100 equal parts.

If stuck: Read the learning goal together.

Slide 93. Recall | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{1}{18}$; stories vary; 2: 24 runners; 3: At 1 $\frac{3}{4}$, the seventh quarter-step from 0; 4: 32 laps

If stuck: Give time to draw before giving a hint.

Slide 94. Recall: answers | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{1}{18}$; stories vary; 24 runners; At 1 $\frac{3}{4}$, the seventh quarter-step from 0; 32 laps

If stuck: Rebuild one missed item with its model.

Slide 95. Launch | 4 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%.

Expect: P1: 8, 20, 40, and 60 counters

If stuck: Ask which amount belongs at 100%.

Slide 96. Explore | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%.

Expect: P2: 30%; P3: 60 stickers

If stuck: Ask which amount belongs at 100%.

Slide 97. The model | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: 100% labels the unknown whole.

If stuck: Ask which amount belongs at 100%.

Slide 98. Required practice | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%.

Expect: P4: 60 minutes; P5: 21 tiles

If stuck: Ask which amount belongs at 100%.

Slide 99. Required practice | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%.

Expect: P6: $2.70

If stuck: Ask which amount belongs at 100%.

Slide 100. Watch the thinking | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 60 blocks

If stuck: Return to the matching unit in the model.

Slide 101. Your turn | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 6 coins; 60 coins

If stuck: Compare with the previous example without copying its numbers.

Slide 102. Compare methods | 5 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%. Partner compare: How does finding 35% of 60 differ from finding a whole when 21 is 35%?

Expect: Accept drawings and equations that preserve the same quantities and units. Combines benchmark percentages.

If stuck: Accept a unit-percent method or a benchmark method.

Slide 103. A claim to check | 5 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: The unknown whole is 60. Check: 40% of 60 is 24.

If stuck: Label 24 at 40% on a double number line and locate 100%.

Slide 104. Words and meaning | 2 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: 35% means 35 for every 100. If all 80 tiles count, 80 is the whole. 25% is one fourth.

If stuck: Read one definition and draw its example.

Slide 105. In your own words | 1 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Use your own words and one example.

Ask: Agree or disagree: The whole must be smaller than 24 if 24 is only 40%.

Expect: Percent compares an amount with a whole split into 100 equal parts.

If stuck: Point to the model and name what each part represents.

Slide 106. Exit ticket | 5 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 60 marbles

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 107. The week ahead | 5 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 108. Mini Check A | 15 min

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Say: Use the printed check and keep the key out of view.

Ask: Read each question before starting.

Expect: Score by skill using the separate answer key.

If stuck: Let the student skip and return. Record unattempted items separately.

Slide 109. Tutor reference

Topic: Percent as a rate per hundred
Student: Day 016, pages 68-78

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 60 marbles

Got it: Finds 60 marbles and checks that 35% of 60 is 21.

Almost there: Sets 21 at 35% correctly but makes a small scaling error or omits the check.

Needs re-teaching: Computes 35% of 21 or gives a whole smaller than the part.

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 017. Monday | pages 79-80

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Ratios and rates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 018. Tuesday | pages 81-82

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Percent practice: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 019. Wednesday | pages 83-84

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 020. Thursday | pages 85-86

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Decimal practice: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 021. Friday | pages 87-88

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Rates and unit rates

Day 015, student workbook pages 63-67

R1 [F-ADD] | student p. 63

Answer: $\frac{11}{12}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

R2 [R-WHOLE] | student p. 63

Answer: 8 yellow; 20 purple

Reasoning: Add ratio parts: 2 + 5 = 7. Divide the total by 7, then multiply each ratio part.

R3 [F-DIV] | student p. 63

Answer: $\frac{1}{6}$ liter

Reasoning: Split the given fractional amount into the stated number of equal shares.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

R4 [R-EQUIV] | student p. 63

Answer: 3 cups

Reasoning: Find the number of recipe batches by dividing the rice amount by 3. Each batch needs 2 cups beans.

S [EXPLAIN] | student p. 63

Answer: Answers vary. A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [R-RATE] | student p. 64

Answer: 5 kilometers per hour

Reasoning: Divide 20 kilometers by 4 hours.

P2 [R-RATE] | student p. 64

Answer: $0.75 per apple

Reasoning: Share the total cost equally among eight apples.

P3 [R-RATE] | student p. 64

Answer: $\frac{2}{5}$ cup per serving

Reasoning: Divide two cups among five servings. The unit is cups per serving.

P4 [R-RATE] | student p. 65

Answer: 5 cards per minute; $\frac{1}{5}$ minute per card

Reasoning: Cards per minute uses 15 ÷ 3. Minutes per card uses 3 ÷ 15.

P5 [R-COMP] | student p. 65

Answer: Pack B: $0.35 per pencil

Reasoning: A costs $0.40 each. B costs $0.35 each, saving $0.05 per pencil.

P6 [R-RATE] | student p. 65

Answer: 9 miles per hour

Reasoning: An hour has three 20-minute groups, so multiply 3 miles by 3.

P7 [R-TABLE] | student p. 66

Answer: 0, 8, and 20 meters

Reasoning: Multiply seconds by 4 meters per second.

F [R-RATE] | student p. 66

Answer: 2; 10 miles per hour

Reasoning: Use the worked example in the slides to check each step.

D [R-RATE] | student p. 66

Answer: No. $6 ÷ 8 apples = $0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

Reasoning: She reverses the requested quantities. Write dollars above the division and apples below it.

E [R-RATE] | student p. 67

Answer: 12 miles per hour; 5 minutes per mile

Reasoning: Nine miles in three 15-minute groups means 3 miles per 15 minutes. Four groups give 12 miles in an hour.

Watch for 5 miles per hour; 12 minutes per mile: The numbers belong to different units; label the quantity per one.

Percent as a rate per hundred

Day 016, student workbook pages 68-72

R1 [F-DIV] | student p. 68

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

R2 [R-EQUIV] | student p. 68

Answer: 24 runners

Reasoning: Eighteen walkers make six groups. Six groups need twenty-four runners.

R3 [F-LINE] | student p. 68

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

R4 [R-TABLE] | student p. 68

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

S [EXPLAIN] | student p. 68

Answer: Answers vary. Percent compares an amount with a whole split into 100 equal parts.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [P-PART] | student p. 69

Answer: 8, 20, 40, and 60 counters

Reasoning: Ten percent is a tenth. Twenty-five percent is a fourth. Fifty percent is half.

P2 [P-PERCENT] | student p. 69

Answer: 30%

Reasoning: $\frac{12}{40} = \frac{3}{10} = \frac{30}{100}$.

P3 [P-WHOLE] | student p. 69

Answer: 60 stickers

Reasoning: Four 10% groups contain 24 stickers. One group has 6; ten groups have 60.

P4 [P-PART] | student p. 70

Answer: 60 minutes

Reasoning: 100% is 48. Another 25% is 12. Add 48 and 12.

P5 [P-PART] | student p. 70

Answer: 21 tiles

Reasoning: 30% is 18 and 5% is 3. Their sum is 21.

P6 [P-PART] | student p. 70

Answer: $2.70

Reasoning: 10% is $1.80. Half of that is 5%, or $0.90. Add them.

F [P-WHOLE] | student p. 71

Answer: 6 coins; 60 coins

Reasoning: Use the worked example in the slides to check each step.

D [P-WHOLE] | student p. 71

Answer: The unknown whole is 60. Check: 40% of 60 is 24.

Reasoning: He treats the known part as the whole. Label 24 at 40% on a double number line and locate 100%.

E [P-WHOLE] | student p. 72

Answer: 60 marbles

Reasoning: Seven 5% groups contain 21 marbles. Each 5% group has 3. Twenty groups have 60.

Watch for 7.35: Taking 35% of 21 finds a part of the given part, not the whole.

Mini Check A

Day 016, student workbook pages 73-74

Score by skill, including the requested reasoning. Use 80% or more as secure, 50-79% as developing, and below 50% as a review priority. Revisit a missed model, then check again with a different problem. This tutoring check does not determine school placement.

1 [F-COMP] | student p. 73

Answer: $\frac{2}{9} < \frac{7}{9}$

Reasoning: With ninths fixed, compare two parts and seven parts.

Watch for $\frac{2}{9} > \frac{7}{9}$: The numerator counts the parts.

2 [F-COMP] | student p. 73

Answer: $\frac{7}{8}$

Reasoning: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

3 [R-RATIO] | student p. 73

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

4 [R-EQUIV] | student p. 73

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

5 [R-RATE] | student p. 73

Answer: 24 miles

Reasoning: Five minutes corresponds to two miles. Twelve five-minute groups make 24 miles.

6 [R-COMP] | student p. 74

Answer: Pack B: $0.40 each

Reasoning: Compare forty-five cents with forty cents for one eraser.

7 [P-PART] | student p. 74

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

8 [P-PART] | student p. 74

Answer: 28 tiles

Reasoning: Thirty percent is 24. Five percent is 4. Add them.

Study guide: Try it

Day 016, student workbook pages 75-78

1 [R-RATE] | student p. 78

Answer: 24 miles

Reasoning: Five minutes corresponds to two miles. Twelve five-minute groups make 24 miles.

2 [R-COMP] | student p. 78

Answer: Pack B: $0.40 each

Reasoning: Compare forty-five cents with forty cents for one eraser.

3 [P-PART] | student p. 78

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

4 [P-PART] | student p. 78

Answer: 28 tiles

Reasoning: Thirty percent is 24. Five percent is 4. Add them.

5 [P-PART] | student p. 78

Answer: 1.8000 dollars

Reasoning: Ten percent is $1.50 and two percent is $0.30. Add them.

Daily practice

Day 017, student workbook pages 79-80

1 [N-FACT] | student p. 79

Answer: 15

2 [N-FACT] | student p. 79

Answer: 35

3 [N-FACT] | student p. 79

Answer: 20

4 [N-FACT] | student p. 79

Answer: 45

5 [R-RATE] | student p. 79

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

6 [R-RATE] | student p. 79

Answer: $\frac{2}{5}$ cups per player; 2 $\frac{1}{2}$ players per cup

Reasoning: Divide the first quantity by the second for cups per player. Reverse for players per cup.

7 [F-SCALE] | student p. 79

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

8 [D-OPS] | student p. 79

Answer: 3.75

9 [R-RATE] | student p. 80 | Built-in review: Ratios and rates

Answer: 24 miles

Reasoning: Five minutes corresponds to two miles. Twelve five-minute groups make 24 miles.

10 [R-COMP] | student p. 80 | Built-in review: Ratios and rates

Answer: Pack B: $0.40 each

Reasoning: Compare forty-five cents with forty cents for one eraser.

11 [R-RATE] | student p. 80 | Built-in review: Ratios and rates

Answer: 4 labels per minute; $\frac{1}{4}$ minute per label

Reasoning: Divide labels by minutes for the first rate. Reverse for the second.

12 [R-TABLE] | student p. 80 | Built-in review: Ratios and rates

Answer: 0, 12, and 21 centimeters

Reasoning: Multiply each time by the unit rate. Zero time matches zero distance.

13 [R-RATE] | student p. 80

Answer: No. $6 ÷ 8 apples = $0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

Watch for $\frac{4}{3}$ dollars per apple: She reverses the requested quantities.

Daily practice

Day 018, student workbook pages 81-82

1 [F-EQUIV] | student p. 81

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 81

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 81

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 81

Answer: 0.125; 12.5%

5 [P-PART] | student p. 81

Answer: 4, 20, and 30 counters

Reasoning: Use one tenth, one half, and three fourths of the whole.

6 [P-PART] | student p. 81

Answer: 1.8000 dollars

Reasoning: Ten percent is $1.50 and two percent is $0.30. Add them.

7 [P-PART] | student p. 81

Answer: 6 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

8 [P-PART] | student p. 81

Answer: 10 stickers

Reasoning: Twenty-five percent is one fourth. Divide the whole by 4.

9 [P-PART] | student p. 82 | Built-in review: Percent practice

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

10 [P-PART] | student p. 82 | Built-in review: Percent practice

Answer: 28 tiles

Reasoning: Thirty percent is 24. Five percent is 4. Add them.

11 [P-WHOLE] | student p. 82 | Built-in review: Percent practice

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

12 [P-PERCENT] | student p. 82 | Built-in review: Percent practice

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

13 [P-WHOLE] | student p. 82

Answer: The unknown whole is 60. Check: 40% of 60 is 24.

Watch for 9.6: He treats the known part as the whole.

Daily practice

Day 019, student workbook pages 83-84

1 [N-SCALE] | student p. 83

Answer: 52

2 [N-SCALE] | student p. 83

Answer: 5

3 [N-SCALE] | student p. 83

Answer: 11

4 [N-SCALE] | student p. 83

Answer: 15

5 [R-RATE] | student p. 83

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

6 [P-PERCENT] | student p. 83

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

7 [R-TABLE] | student p. 83

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

8 [F-LINE] | student p. 83

Answer: Halfway

Reasoning: The interval is 3 units. One and one half units is half of that interval.

9 [F-ADD] | student p. 84

Answer: $\frac{1}{2}$

10 [N-SCALE] | student p. 84

Answer: 5

11 [N-FACT] | student p. 84

Answer: 24

12 [G-VOL] | student p. 84

Answer: 72 cm^3

13 [PUZZLE] | student p. 84 | Built-in review: Weekly puzzle

Answer: 40

Reasoning: The gap is 15%, which equals 6. Five percent is 2, so 100% is 40.

Daily practice

Day 020, student workbook pages 85-86

1 [N-EST] | student p. 85

Answer: 500

2 [N-EST] | student p. 85

Answer: 6

3 [N-EST] | student p. 85

Answer: About 100

4 [N-EST] | student p. 85

Answer: About 20

5 [R-RATE] | student p. 85

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

6 [P-PERCENT] | student p. 85

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

7 [P-WHOLE] | student p. 85

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

8 [F-DIV] | student p. 85

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

9 [D-OPS] | student p. 86 | Built-in review: Decimal practice

Answer: 4.15

10 [D-OPS] | student p. 86 | Built-in review: Decimal practice

Answer: 1.70

11 [D-OPS] | student p. 86 | Built-in review: Decimal practice

Answer: 6.15

12 [D-OPS] | student p. 86 | Built-in review: Decimal practice

Answer: 2.70

13 [R-RATE] | student p. 86

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

14 [P-WHOLE] | student p. 86

Answer: The unknown whole is 60. Check: 40% of 60 is 24.

Watch for 9.6: He treats the known part as the whole.

Weekly check

Day 021, student workbook pages 87-88

1 [R-RATE] | student p. 87

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

2 [P-PERCENT] | student p. 87

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

3 [P-WHOLE] | student p. 87

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

4 [R-RATE] | student p. 87

Answer: 4 miles per hour

Reasoning: Use the named units, calculate, then check with the inverse operation.

5 [F-SCALE] | student p. 87

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

6 [F-DIV] | student p. 88

Answer: 12 pieces

Reasoning: Each meter contains 4 equal pieces. Multiply by 3 meters.

Watch for $\frac{3}{4}$: Multiplying by the divisor answers a different question.

7 [R-RATIO] | student p. 88

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

8 [F-ADD] | student p. 88

Answer: $\frac{1}{3}$

9 [N-SCALE] | student p. 88

Answer: 7

10 [R-RATE] | student p. 88

Answer: No. \$6 ÷ 8 apples = \$0.75 per apple. $\frac{4}{3}$ describes apples per dollar.

Watch for $\frac{4}{3}$ dollars per apple: She reverses the requested quantities.